

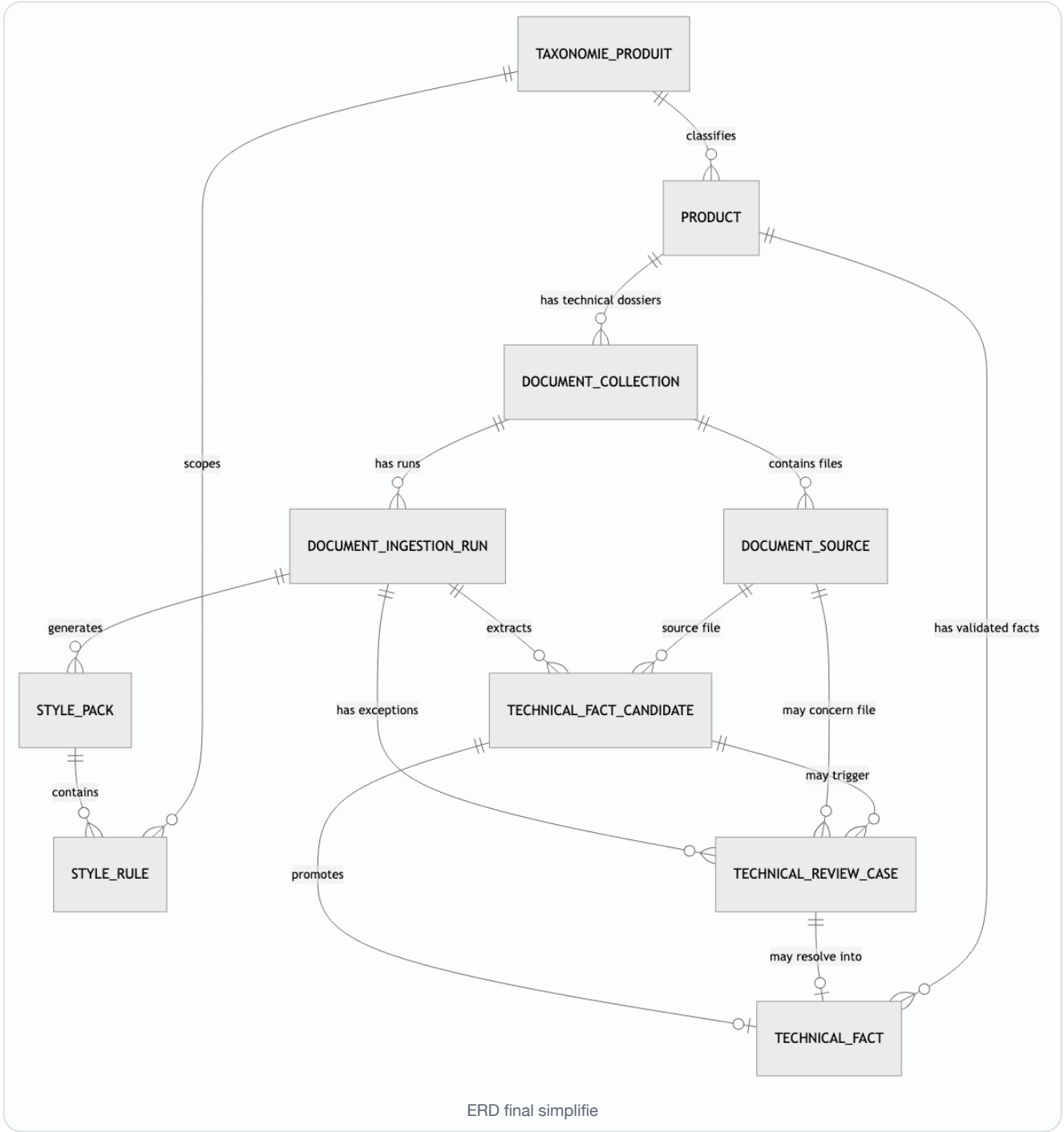
ERD POC final simplifie

Ce document fige le modele POC simplifie pour l'ingestion du guide de style et des dossiers techniques usine.

Le perimetre POC est volontairement limite :

- style guide : un dossier global, un PDF, extraction via Document AI Layout Parser, generation d'un pack candidat, validation humaine regle par regle ;
- dossiers techniques usine : un dossier par produit, plusieurs PDFs separees, Custom Classifier, Enterprise OCR, Custom Extractor Foundation Model, validation deterministe et review humaine par exception ;
- pas de Custom Splitter ;
- pas de table de preuves abstraite ;
- une preuve principale denormalisee dans `style_rule` et `technical_fact_candidate`.

ERD final simplifie



Enums simplifies

Enum	Valeurs
collection_kind	STYLE_GUIDE , TECHNICAL_DOSSIER
document_type	STYLE_GUIDE , TECHNICAL_SHEET , BLUEPRINT , ECO_CERTIFICATE , ASSEMBLY_NOTICE , MATERIAL_SPECIFICATION , UNKNOWN
document_collection.statut	EN_ATTENTE , EN_COURS , A_VALIDER , TERMINE , ERREUR
document_source.statut	EN_ATTENTE , EN_COURS , TERMINE , ERREUR
document_ingestion_run.statut	EN_ATTENTE , EN_COURS , A_VALIDER , TERMINE , ERREUR , ANNULE
current_step	UPLOAD , DOCUMENT_CLASSIFICATION , OCR_PROOF , LAYOUT_PARSE , FACT_EXTRACTION , DETERMINISTIC_VALIDATION , LLM_DRAFT_PACK , HUMAN_REVIEW , PROMOTION , DONE
style_pack.statut	BROUILLON , ACTIF , ARCHIVE
style_rule.decision_editoriale	A_VALIDER , APPROUVEE , DESACTIVEE
style_rule.origine	LLM , HUMAIN , MODIFIEE
type_regle	VOIX , TON , FORMATAGE , PROMESSE_INTERDITE
niveau_contrainte	HARD , SOFT
technical_fact_candidate.validation_status	AUTO_VALIDATED , NEEDS_REVIEW , REJECTED , PROMOTED
technical_review_case.case_type	CLASSIFICATION_UNCERTAIN , LOW_OCR_QUALITY , DOCUMENT_UNREADABLE , MISSING_REQUIRED_FIELD , LOW_CONFIDENCE , VALUE_OUT_OF_RANGE , EXACT_MATCH_FAILED , CONTRADICTION , LLM_SELF_CHECK_FLAG
technical_review_case.trigger_source	CLASSIFIER , OCR , CUSTOM_EXTRACTOR , PYTHON_VALIDATOR , LLM_SELF_CHECK
technical_review_case.severity	BLOCKING , WARNING
technical_review_case.status	A_TRAITER , APPROUVE , CORRIGE , REJETE , DOCUMENT_A_REEMPLACER
technical_review_case.resolution_action	APPROVE_DETECTED_VALUE , CORRECT_VALUE , REJECT_VALUE , REQUEST_NEW_DOCUMENT
technical_fact.validation_source	SYSTEM , HUMAN
extraction_method	EXTRACT , DERIVE

1. product

Champ	Exemple complet
id	prod_001
sku	AXO-TABLE-TECK-190
name	Table en teck 190 cm
taxonomie_produit_id	taxo_mobilier_jardin
created_at	2026-04-20T10:00:00Z
updated_at	2026-04-20T10:00:00Z

2. document_collection

Champ	Exemple style guide	Exemple dossier usine
id	coll_style_001	coll_tech_001
collection_kind	STYLE_GUIDE	TECHNICAL_DOSSIER
product_id	null	prod_001
statut	A_VALIDER	A_VALIDER
replaced_by_collection_id	null	null
dernier_message_erreur	null	Certification FSC manquante
created_at	2026-04-20T10:00:00Z	2026-04-20T10:10:00Z
updated_at	2026-04-20T10:30:00Z	2026-04-20T10:30:00Z

Contraintes :

```
STYLE_GUIDE => product_id IS NULL
TECHNICAL_DOSSIER => product_id IS NOT NULL
```

3. document_source

Champ	Exemple style guide	Exemple dossier usine
id	src_style_pdf_001	src_tech_sheet_001
collection_id	coll_style_001	coll_tech_001
original_file_name	AXOLOTL_STYLE_GUIDE_V4.pdf	fiche_technique.pdf
storage_uri	gs://factory-writer/style-guide.pdf	gs://factory-writer/fiche_technique.pdf
storage_bucket	factory-writer	factory-writer
storage_object_name	sources/style-guides/style-guide.pdf	technical-dossiers/prod_001/fiche_technique.pdf
storage_generation	1776626134167000	1776627000000000
storage_metageneration	1	1
storage_content_type	application/pdf	application/pdf
storage_size_bytes	842139	1934021
document_type	STYLE_GUIDE	TECHNICAL_SHEET
classification_confidence	null	0.94
quality_score_min	null	0.82
quality_score_avg	null	0.93
quality_metadata_json	null	{"defects":[]}
statut	TERMINE	TERMINE
replaced_by_source_id	null	null
dernier_message_erreur	null	null
created_at	2026-04-20T10:00:00Z	2026-04-20T10:10:00Z
updated_at	2026-04-20T10:20:00Z	2026-04-20T10:20:00Z

4. document_ingestion_run

Champ	Exemple style guide	Exemple dossier usine
id	run_style_001	run_tech_001
collection_id	coll_style_001	coll_tech_001
pipeline_kind	STYLE_GUIDE_EXTRACTION	TECHNICAL_DOSSIER_EXTRACTION
statut	A_VALIDER	A_VALIDER
current_step	HUMAN_REVIEW	HUMAN_REVIEW
temporal_workflow_id	style-guide-ingestion-run_style_001	technical-dossier-ingestion-run_tech_001
temporal_run_id	7c1a...	8d2b...
extraction_steps_json	voir exemple ci-dessous	voir exemple ci-dessous
validation_summary_json	{"rules_generated":12,"rules_to_review":12}	{"auto_validated":8,"review_cases":2}
error_message	null	null
started_at	2026-04-20T10:02:00Z	2026-04-20T10:15:00Z
completed_at	null	null
created_at	2026-04-20T10:02:00Z	2026-04-20T10:15:00Z
updated_at	2026-04-20T10:28:00Z	2026-04-20T10:28:00Z

Exemple extraction_steps_json style guide :

```
[
  {
    "step_kind": "LAYOUT_PARSE",
    "provider": "google_document_ai",
    "processor_kind": "layout_parser",
    "processor_version": "pretrained-layout-parser-v1.6-pro-2025-12-01",
    "provider_job_id": "projects/.../operations/123",
    "status": "SUCCEEDED",
    "output_uri": "gs://.../style-guide-layout/run_style_001/",
    "request_config": {
      "chunking_config": {
        "chunk_size": 1000,
        "include_ancestor_headings": true
      }
    }
  },
  {
    "step_kind": "LLM_DRAFT_PACK",
    "provider": "litellm_sdk",
    "model": "vertex_ai/gemini-3-pro-preview",
    "status": "SUCCEEDED"
  }
]
```

Exemple extraction_steps_json dossier usine :

```
[
  {
    "step_kind": "DOCUMENT_CLASSIFICATION",
    "provider": "google_document_ai",
    "processor_kind": "custom_classifier",
    "processor_version": "classifier-v1",
    "status": "SUCCEEDED"
  },
  {
    "step_kind": "OCR_PROOF",
    "provider": "google_document_ai",
    "processor_kind": "enterprise_ocr",
    "request_config": {
      "enableNativePdfParsing": true,
      "enableImageQualityScores": true
    },
    "status": "SUCCEEDED"
  },
  {
    "step_kind": "FACT_EXTRACTION",
    "provider": "google_document_ai",
    "processor_kind": "custom_extractor_foundation_model",
    "processor_version": "foundation-v1",
    "status": "SUCCEEDED"
  },
  {
    "step_kind": "DETERMINISTIC_VALIDATION",
    "provider": "factory_writer",
    "status": "NEEDS_REVIEW"
  }
]
```

5. style_pack

Champ	Exemple complet
id	pack_001
ingestion_run_id	run_style_001
statut	BROUILLON
est_actif	false
prompt_registry_provider	local
prompt_name	style_guide_extract_rules
prompt_version	v1
llm_model	vertex_ai/gemini-3-pro-preview
llm_temperature	0.0
llm_max_tokens	8192
llm_response_format_name	style_pack_candidate_v1
rendered_system_prompt_hash	sha256:abc...
rendered_user_prompt_hash	sha256:def...
validation_summary_json	{"rules_validated":12,"deduplicated":1}
approuve_le	null
created_at	2026-04-20T10:05:00Z
updated_at	2026-04-20T10:05:00Z

6. `style_rule`

Champ	Exemple complet
<code>id</code>	<code>rule_001</code>
<code>pack_id</code>	<code>pack_001</code>
<code>taxonomie_produit_id</code>	<code>taxo_mobilier_jardin</code>
<code>type_regle</code>	<code>TON</code>
<code>niveau_contrainte</code>	<code>SOFT</code>
<code>texte_regle_original</code>	<code>Favoriser matière, stabilité et scène extérieure.</code>
<code>texte_regle</code>	<code>Favoriser la matière, la stabilité et l'usage extérieur.</code>
<code>decision_editoriale</code>	<code>A_VALIDER</code>
<code>est_actif</code>	<code>false</code>
<code>origine</code>	<code>LLM</code>
<code>source_evidence_text</code>	<code>SR-01 mobilier_jardin Favoriser matière, stabilité, confort d'usage et scène de vie extérieure.</code>
<code>source_evidence_provider_id</code>	<code>c1</code>
<code>source_evidence_page_start</code>	<code>1</code>
<code>source_evidence_page_end</code>	<code>1</code>
<code>source_evidence_json</code>	<code>{"chunkId":"c1","pageSpan":{"pageStart":1,"pageEnd":1}}</code>
<code>commentaire_review</code>	<code>null</code>
<code>reviewed_at</code>	<code>null</code>
<code>reviewed_by</code>	<code>null</code>
<code>created_at</code>	<code>2026-04-20T10:05:10Z</code>
<code>updated_at</code>	<code>2026-04-20T10:05:10Z</code>

7. `technical_fact_candidate`

Champ	Exemple complet
<code>id</code>	<code>fact_cand_001</code>
<code>ingestion_run_id</code>	<code>run_tech_001</code>
<code>source_id</code>	<code>src_tech_sheet_001</code>
<code>field_name</code>	<code>dimension_width_cm</code>
<code>raw_value</code>	<code>190 cm</code>
<code>normalized_value</code>	<code>190</code>
<code>unit</code>	<code>cm</code>
<code>extractor_confidence</code>	<code>0.96</code>
<code>extraction_method</code>	<code>EXTRACT</code>
<code>validation_status</code>	<code>AUTO_VALIDATED</code>
<code>review_required</code>	<code>false</code>
<code>review_reason</code>	<code>null</code>
<code>source_evidence_text</code>	<code>190 cm</code>
<code>source_page</code>	<code>2</code>
<code>source_bbox_json</code>	<code>{"left":0.12,"top":0.43,"width":0.04,"height":0.02}</code>
<code>raw_entity_json</code>	<code>{"type":"dimension_width_cm","mentionText":"190 cm","confidence":0.96}</code>
<code>created_at</code>	<code>2026-04-20T10:24:00Z</code>
<code>updated_at</code>	<code>2026-04-20T10:24:00Z</code>

8. `technical_review_case`

Champ	Exemple complet
<code>id</code>	<code>case_001</code>
<code>ingestion_run_id</code>	<code>run_tech_001</code>
<code>source_id</code>	<code>src_tech_sheet_001</code>
<code>fact_candidate_id</code>	<code>fact_cand_001</code>
<code>case_type</code>	<code>VALUE_OUT_OF_RANGE</code>
<code>trigger_source</code>	<code>PYTHON_VALIDATOR</code>
<code>severity</code>	<code>BLOCKING</code>
<code>status</code>	<code>A_TRAITER</code>
<code>field_name</code>	<code>dimension_width_cm</code>
<code>title</code>	<code>Largeur hors bornes attendues</code>
<code>description</code>	<code>La valeur 1900 cm dépasse la borne maximale attendue pour mobilier_jardin.</code>
<code>detected_value</code>	<code>1900</code>
<code>detected_unit</code>	<code>cm</code>
<code>suggested_value</code>	<code>190</code>
<code>suggested_unit</code>	<code>cm</code>
<code>corrected_value</code>	<code>null</code>
<code>corrected_unit</code>	<code>null</code>
<code>resolution_action</code>	<code>null</code>
<code>resolution_comment</code>	<code>null</code>
<code>resolved_fact_id</code>	<code>null</code>
<code>resolved_by</code>	<code>null</code>
<code>resolved_at</code>	<code>null</code>
<code>metadata_json</code>	<code>{"bounds":{"min":40,"max":400}}</code>
<code>created_at</code>	<code>2026-04-20T10:25:00Z</code>
<code>updated_at</code>	<code>2026-04-20T10:25:00Z</code>

9. `technical_fact`

Champ	Exemple complet
<code>id</code>	<code>fact_001</code>
<code>product_id</code>	<code>prod_001</code>
<code>source_candidate_id</code>	<code>fact_cand_001</code>
<code>field_name</code>	<code>dimension_width_cm</code>
<code>value</code>	<code>190</code>
<code>unit</code>	<code>cm</code>
<code>validation_source</code>	<code>SYSTEM</code>
<code>validated_at</code>	<code>2026-04-20T10:26:00Z</code>
<code>validated_by</code>	<code>system</code>
<code>created_at</code>	<code>2026-04-20T10:26:00Z</code>
<code>updated_at</code>	<code>2026-04-20T10:26:00Z</code>

10. `taxonomie_produit`

Champ	Exemple complet
<code>id</code>	<code>taxo_mobilier_jardin</code>
<code>famille_code</code>	<code>mobilier_jardin</code>
<code>libelle_fr</code>	<code>Mobilier de jardin</code>
<code>parent_id</code>	<code>null</code>
<code>created_at</code>	<code>2026-04-20T09:00:00Z</code>
<code>updated_at</code>	<code>2026-04-20T09:00:00Z</code>

Contraintes cles

```
document_collection.collection_kind = STYLE_GUIDE
=> product_id IS NULL

document_collection.collection_kind = TECHNICAL_DOSSIER
=> product_id IS NOT NULL

product.taxonomie_produit_id
=> FK taxonomie_produit.id

document_source.storage_bucket + storage_object_name + storage_generation
=> unique

document_ingestion_run.temporal_workflow_id
=> unique

style_pack.est_actif = true
=> unique global pour le POC

technical_fact.product_id + field_name
=> unique

technical_review_case.resolved_fact_id
=> nullable, FK technical_fact.id

technical_review_case.fact_candidate_id
=> nullable, car certains cas n'ont aucun fact candidat

technical_review_case.source_id
=> nullable, car certains cas peuvent etre globaux au dossier
```

Positionnement POC

Ce modele garde :

- le guide de style ;
- les dossiers techniques multi-PDF ;
- la review humaine ;
- les faits valides ;
- les preuves principales ;
- le lien produit-taxonomie.

Il evite volontairement :

- une table de preuves abstraite ;
- une table de preuves style-rule ;
- le Custom Splitter ;
- une table `document_extraction_step` separee.

Les details providers et configs restent dans `document_ingestion_run.extraction_steps_json`.